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# Structure-Feature Co-Optimization for Dissolved Gas Analysis in Power Transformer

Jinpeng Yang, Huifang Ma, Jinfa Han, Rui Bing, and Meihuizi Jia

***Abstract*— Dissolved Gas Analysis (DGA) is a fundamental technique for diagnosing faults in oil-immersed power transformers. Although graph neural networks have improved DGA modeling, they still face two major challenges: (1) *low-quality graphs* and (2) *high inter-class similarity*. Poor graph construction introduces structural noise and bias, causing misleading message propagation and semantic misalignment, which degrades node representations and prediction accuracy. Meanwhile, high inter-class similarity leads to feature redundancy and weak discriminability. To overcome these challenges, we propose STructurEfeAture Co-OptiMization (STEAM) for DGA in power transformer, a novel framework that jointly performs topology optimization and feature denoising. STEAM integrates three key components: (1) the Edge Reconstruction with Structural Consistency (ERSC) module reconstructs the topology of the graph using feature similarity and label information; (2) The Joint Optimization with Structure-Feature Consistency (JOSFC) module uses graph feature spectrum alignment and Singular Value Decomposition (SVD) denoising to collaborate with ERSC to construct a purer fault relationship graph to address the challenge of weak graph quality; (3) The Class-wise Prototype Alignment (CPA) module replaces sample level comparison with category prototype comparison to further enhance discriminative power to address the challenge of high inter-class similarity. Extensive experiments demonstrate that our method outperforms existing state-of-the-art approaches, achieving performance improvements of 1.95%–4.94% on various metrics on the DGA-2000 dataset and 0.81%–2.3% on the DGA-200 dataset.**

***Index Terms*— Dissolved gas analysis, Graph neural networks, Topology optimization, Feature denoising, Prototype alignment.**

## I. Introduction

As a key component of modern power systems, the reliability of power transformers directly affects grid safety and stability. Effective condition monitoring and fault diagnosis are therefore essential [1]. Dissolved Gas Analysis (DGA) is widely used for preventive maintenance of oil-immersed transformers by detecting the types (e.g., $H_2$, $CH_4$, $C_2H_6$, $C_2H_4$, $C_2H_2$) [2] and concentrations of dissolved gases, enabling diagnosis of latent faults such as partial discharge (PD), spark discharge (SD), arc discharge (AD), low- to high-temperature overheating (LT, LMT, MT, HT), high- and low-energy discharges (HE, LE), while also identifying normal type, thus facilitating timely interventions to prevent catastrophic failures.

Traditional DGA diagnostic methods [3] have been widely adopted. These methods rely on expert-defined rules and thresholds, which are static and may not generalize well to complex or evolving operational conditions, such as changes in load, temperature, and insulation aging. In contrast, machine learning-based approaches [3], [4] learn diagnostic models directly from data, offering the potential to capture non-linear patterns and achieve more robust performance under diverse operating conditions. However, most existing methods assume that samples are independent and identically distributed (i.i.d.) and primarily focus on modeling feature–label relationships, without explicitly capturing structured inter-sample relationships (e.g., similarity or topology). It is important to note that both types of methods, once implemented, can be deployed in automated monitoring systems when gas data are available—the main practical bottlenecks for real-time application lie in the upstream stages of oil sampling, gas extraction, and quantification.These models typically ignore the intrinsic relationships between samples and are highly sensitive to noise. In real-world DGA scenarios, data uncertainty remains a critical challenge. Noise may arise from multiple sources, including variations in oil sampling procedures, errors in gas extraction and chromatographic measurements, fluctuations in operating conditions (such as load and temperature), and potential ambiguities in fault labeling when multiple fault mechanisms coexist. Although preprocessing techniques have been proposed in prior studies to mitigate noise, residual uncertainty may still degrade diagnostic performance. To address these limitations, graph-based learning paradigms have recently gained attention, as they enable explicit modeling of inter-sample relationships and provide a flexible framework for improving robustness under complex conditions.

Initially proposed by [5], Graph Neural Networks (GNNs) have seen rapid advancements in recent years, with typical models including Graph Convolutional Networks [6], Graph Attention Networks (GAT) [7], and GraphSAGE [8]. Owing to their strong ability to capture relational inductive biases, an increasing number of tasks can be effectively addressed using these models. The DGA task is no exception. Recent graph-based approaches [9], [10] address fault diagnosis by first transforming tabular DGA data into a graph structure, where each oil sample is represented as a node associated with a feature vector of gas concentrations. Edges between nodes are constructed based on feature similarity—typically via k-nearest neighbors (KNN) or distance thresholds—to

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capture both homophilic and heterophilic relationships among samples. This dual construction is crucial: homophilic edges facilitate the aggregation of intra-class information, reinforcing consistent features within the same fault type, while heterophilic edges help preserve discriminative boundaries between different fault categories, preventing over-smoothing in GNN message passing. By jointly exploiting these complementary interactions, GNNs propagate and refine diagnostic information across the graph, learning node representations that encode both local feature patterns and global fault dependencies.

In this work, we focus on the modeling stage given an existing labeled dataset and do not address uncertainties introduced during upstream processes such as oil sampling, gas extraction, and chromatographic measurement. Our objective is to improve diagnostic performance by better exploiting the structural relationships among samples. Although graph-based methods have improved diagnostic performance, their effectiveness heavily depends on graph quality and representation separability. Two major challenges persist: (1) Weak Graph Quality. DGA data is inherently tabular, and constructing graphs by treating samples as nodes can introduce edges between different fault categories, violating the homophily assumption essential for GNN message passing [11]. This issue is further exacerbated when feature corruption—such as measurement inaccuracies or labeling ambiguities—distorts sample representations, leading to unreliable graph structures. Balancing noise suppression with preservation of critical fault signatures remains difficult. (2) High Inter-Class Similarity. Statistical analysis of benchmark datasets (DGA-2000[1] and DGA-200[2], see Section IV.A) reveals high inter-class similarity (Fig. 1), leading to blurred decision boundaries. This challenge is rooted in the intrinsic characteristics of fault evolution—different fault types (e.g., MT vs. HT) can exhibit similar gas patterns, resulting in strong inter-class correlations and complex non-linear relationships among samples. Traditional losses like cross-entropy often fail to enforce intra-class compactness and inter-class separability under such conditions. Most existing methods address either topological refinement [12] or feature denoising [13] in isolation, neglecting the coupling between graph structure and features representations. Consequently, they yield suboptimal graph quality and discriminability, particularly when inter-class similarity is high. This motivates our proposed approach, which jointly optimizes both aspects to enhance intra-class compactness and inter-class separability.

Building upon the analysis of the two intertwined challenges above, we argue that the noise in graph structure and node features is intrinsically coupled. Consequently, methods that target only one modality, whether topological rewiring that overlooks feature corruption, or spectral alignment that remains vulnerable to an initially heterophilic structure, inevitably yield suboptimal gains when deployed in isolation. In response, we propose **STEAM** (StructurE–feAture Co-OptiMization),

[1]The DGA-2000 is a publicly available dataset for DGA fault diagnosis contains 1982 samples and includes 6 types of faults.

[2]The DGA-200 is a small-scale DGA fault diagnosis dataset contains 201 samples and covers 7 types of faults.

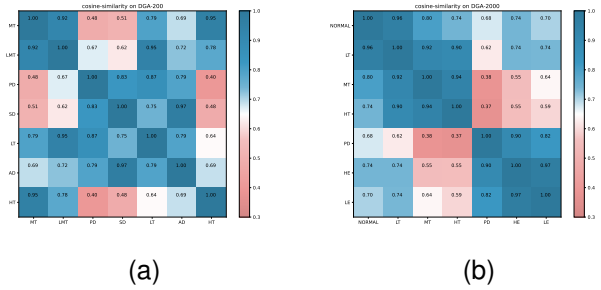



Fig. 1. Sample class center similarity. (a) Results on DGA-200. (b) Results on DGA-2000.

a dual-channel collaborative purification framework tailored for DGA fault diagnosis. Rather than processing structure and features sequentially, STEAM enables them to be jointly and interactively denoised through three complementary modules: (1) Edge Reconstruction with Structural Consistency (**ERSC**) rewires the graph to enhance structural consistency and address structural noise, (2) Joint Optimization with Structure-Feature Consistency (**JOSFC**) synchronizes the graph topology and node features in the spectral domain to filter out feature-structure misalignment and irrelevant feature noise, and (3) Class-wise Prototype Alignment (**CPA**) aligns class-wise prototypes across channels to reinforce semantic consistency, widen inter-class separation, and harden decision boundaries. Crucially, these modules operate in parallel and their outputs are continuously harmonized rather than simply concatenated. This synergistic workflow ensures that structural refinement informs feature denoising and vice versa, thereby generating a high-fidelity, semantically coherent, and discriminative graph representation. By collaboratively optimizing both graph quality and feature discriminability, STEAM establishes an optimal foundation for precise fault classification.

In summary, our core contributions are threefold:

**Problem Formalization**: Identify two coupled challenges in graph-based DGA: weak graph quality and high inter-class similarity.

**Technical Innovation**: Propose STEAM, the first dual-graph collaborative optimization framework that jointly refines structure and features via parallel ERSC and JOSFC channels, enhanced by cross-channel prototype alignment (CPA) for discriminative learning.

**Experimental Validation**: STEAM outperforms state-of-the-art GNNs (GCN, DHGR, JDR) by 3%–10% on DGA-2000 and DGA-200, demonstrating superior robustness and generalization.

The rest of this paper is organized as follows: Preliminaries and important notations are specified in Section II. In Section III, the proposed STEAM framework is presented in detail,including its dual-channel graph enhancement mechanism and class-wise prototype alignment. Extensive experimental results and analysis are provided in Section IV to validate the framework's effectiveness. Section V provides a comprehensive discussion. Finally, Section VI concludes the paper and outlines future research directions.

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## II. PRELIMINARIES

### A. Notation and Problem Formulation

In this work, we model DGA samples as graph-structured data, defined as follows. Let $\mathbf{X} \in \mathbb{R}^{N\times d}$ denote the original feature matrix of N transformer oil samples obtained from DGA, where each sample is represented by a d-dimensional feature vector $\mathbf{x}_i$. Then we construct an original graphs based on $\mathbf{X}$: $G = (V, E_0, \mathbf{X}, \mathbf{A}_0)$ as detailed in Section III.A. The node set $V = \{v_1, v_2, v_3, ..., v_i\}$ represents the set of samples with gas data in the oil. $E_0 \subseteq V \times V$ represents the set of edges, typically established based on feature similarity (e.g., using KNN). The adjacency matrix of the graph is denoted by $\mathbf{A}_0 \in \mathbb{R}^{N\times N}$ , $\mathbf{A}_{0_{ij}}$ where = 1 if edge $e_{i,j} \in E_0$, and 0 otherwise. To account for self-connections during message passing, we augment $\mathbf{A}_0$ with self-loops, obtaining $\hat{\mathbf{A}} = \mathbf{A}_0 + \mathbf{I}_N$, where $\mathbf{I}_N$ is the identity matrix. The symmetrically normalized adjacency matrix is then given by $\tilde{\mathbf{A}} = \hat{\mathbf{D}}^{-1/2}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-1/2}$, where $\hat{\mathbf{D}}$ is the degree matrix of $\hat{\mathbf{A}}$ with $\hat{\mathbf{D}}_{ii} = \sum_j \hat{\mathbf{A}}_{ij}$.

The ground-truth labels are given by $\mathbf{Y} \in \mathbb{R}^{N\times C}$, where $C$ is the number of fault classes. If node $v_i$ belongs to class $c$, then $\mathbf{Y}_{i,c} = 1$ and all other entries in row $i$ are zero. For training, we have access to labels for a subset of nodes $V^L \subset V$; the remaining nodes are unlabeled and used for testing.

The task of transformer fault diagnosis can be formally stated as follows:

**Input:** An initial feature matrix $\mathbf{X}$, an initial adjacency matrix $\mathbf{A}_0$ (constructed from $\mathbf{X}$ via KNN), and ground-truth labels $\mathbf{Y}^L$ corresponding to the training nodes in $V^L$.

**Output:** A learned prediction function $f$ that, for any test sample, produces a probability vector $\hat{\mathbf{y}}_i$ over the $C$ fault classes.

### B. Graph Neural Networks

GNNs learn node representations by propagating and aggregating information from neighboring nodes. Under the message-passing paradigm, GNNs can be defined by stacking two fundamental operations [14]:

**Propagation (P)**: This operation disseminates node information through the graph structure to enhance feature smoothness among connected nodes.

**Transformation (T)**: This operation applies a non-linear transformation to node features to increase expressiveness and mitigate excessive smoothness. The corresponding formula form is as follows:

$$\text{Propagation}(\mathbf{X}) = P(\mathbf{X}) = \tilde{\mathbf{A}} \cdot \mathbf{X}, \tag{1}$$

$$\text{Transformation}(\mathbf{X}) = T(\mathbf{X}) = \sigma(\mathbf{W}\mathbf{X}), \tag{2}$$

$$\text{GCN}(\mathbf{X}) = T(P(T(P(\mathbf{X})))) . \tag{3}$$

where $\sigma(\cdot)$ denotes a non-linear activation function (e.g., ReLU), and $\mathbf{W} \in \mathbb{R}^{d'\times d}$ is a learnable weight matrix that projects input features from dimension $d$ to $d'$. In contrast, disentangled GNNs (e.g., P-P-T-T or T-T-P-P) execute all propagation steps prior to transformation, or vice versa. For instance, the Simplified Graph Convolution (SGC) model [15] is a disentangled (P-P-T-T) architecture formulated as:

$$\text{SGC}(\mathbf{X}) = T(T(P(P(\mathbf{X})))) , \tag{4}$$

where T is executed after all P are finished.

By performing two consecutive transformations before propagation, the initial features of nodes can be compressed, reconstructed, and discriminative enhanced. In the subsequent propagation steps, the input information has been purified, reducing the interference of noise and redundant features. Therefore, we chose the TTPP architecture as the backbone of STEAM.

## III. METHODOLOGY

This section presents the proposed graph-rewired contrastive learning framework for DGA. A similarity-based graph is first constructed from raw samples, then dual-channel graph rewiring via ERSC and JOSFC is performed in parallel to enhance topological quality and address weak structural connectivity. At the same time, JOSFC applies SVD to denoise features. The denoised features are propagated through GNNs on both graphs. Finally, a contrastive prototype alignment module reinforces intra-class compactness and inter-class separation by drawing samples toward their class prototypes and pushing different prototypes apart. The full architecture is shown in Fig. 2.

### A. Graph Construction

To leverage GNNs for DGA, we construct a sample-level graph where nodes represent DGA samples with features from transformer oil gas concentrations. Edges are formed based on feature similarity using a K-nearest neighbors (KNN) strategy [16] to capture class-specific distribution patterns [9], [10].

Specifically, for an original gas node $v_i$, we calculate its similarity with all other nodes to form a similarity vector $s_i$, where each element $s_{ij}$ denotes the similarity between nodes $v_i$ and $v_j$. Repeating this process for every node yields the sorted similarity matrix $\mathbf{S} \in \mathbb{R}^{N\times N}$ with its entry $s_{ij}$ denotes the similarity value between sample $v_i$ and $v_j$. This matrix is subsequently used to construct the KNN graph. For each node $v_i$, we select the top $K$ nodes that are most similar to it to construct a neighbor set $N_i = \arg_{v_j} \text{topk}_{i\neq j}\, s_{ij}$. The adjacency matrix $\mathbf{A}_{0_{ij}}$ is defined as:

$$\mathbf{A}_{0_{ij}} = \begin{cases} 1, & v_j \in N_i \\ 0, & \text{others} \end{cases} , \tag{5}$$

where $\mathbf{A}_{0_{ij}} = 1$ indicates that node $v_j$ is among the $K$ most similar neighbors of node $v_i$. It is worth noting that the choice of the hyperparameter $K$ used to construct the adjacency matrix can significantly influence the final performance of the model, since the structural properties of the graph directly affect representation learning. A detailed discussion on how to select an appropriate value of $K$ is provided in Section IV.A.

### B. Dual-Channel Graph Enhancement

Graph-based modeling for DGA is challenged by weak graph quality due to redundant or missing edges and noisy

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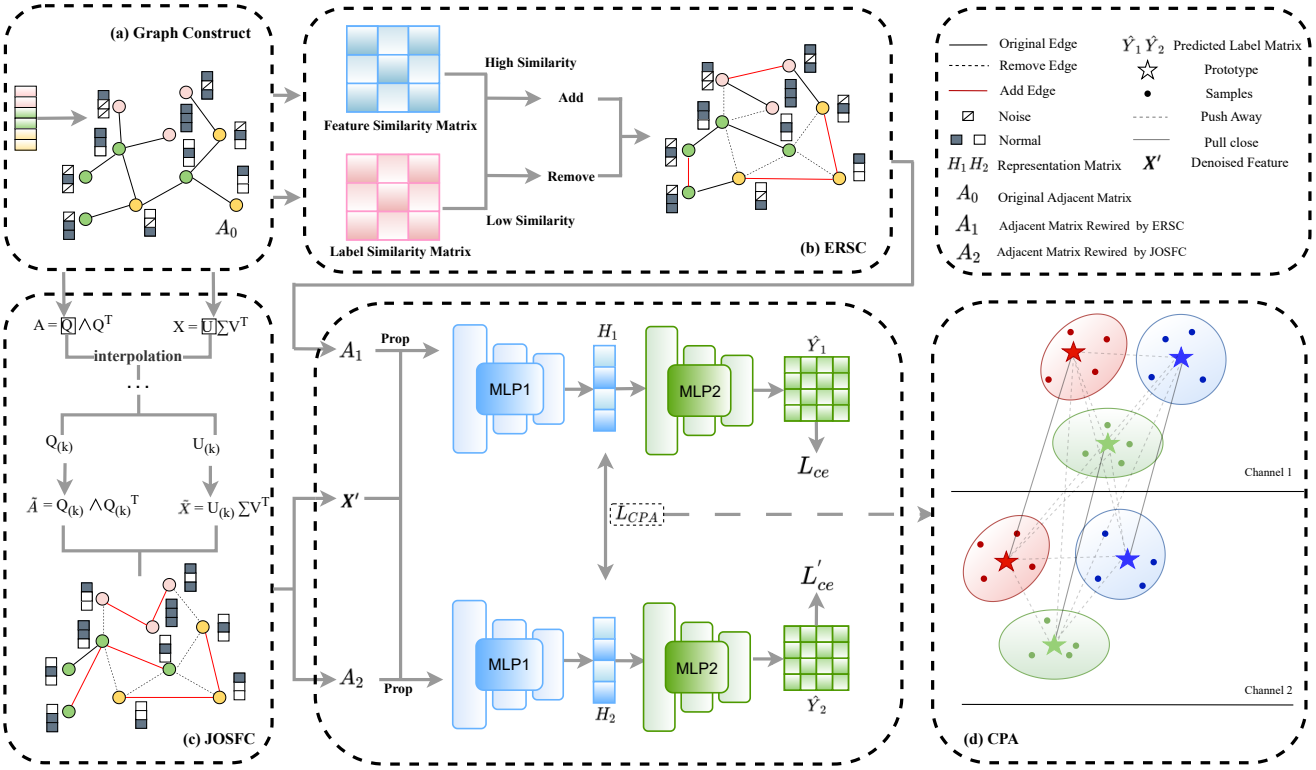



Fig. 2. Overall framework of STEAM. (a) Graph constructed by computing pairwise feature similarities and using a KNN strategy to connect each sample to its top-k most similar samples. (b) ERSC rewires the graph using feature and label similarity. (c) JOSFC captures spectral alignment between graph structure and node features while denoising features and optimizing topology. (d) CPA aligns same-class sample embeddings to prototypes in the embedding space to expand inter-class distances. Each node represents a transformer sample, and nodes with different colors correspond to different fault categories. Edges indicate the structural relationships constructed among samples.

features that obscure true semantics, resulting in unstable representations and amplified errors during GNN message passing.

To address these issues, we propose a dual-channel graph optimization framework. Channel ERSC and Channel JOSFC independently enhance the original adjacency matrix $\mathbf{A}_0$ into $\mathbf{A}_1$ and $\mathbf{A}_2$ by focusing on local structural consistency and structure-feature alignment, respectively. Simultaneously, the framework denoises features to produce $\hat{x}$, providing a firm foundation for dual-channel propagation and improved discriminative graph representations.

*1) ERSC:* To address the challenge of structural heterophily, stemming from complex sample distributions and ambiguous class boundaries in DGA data , we introduce a structural optimization channel named **ERSC**. This module aims to adjust the connection patterns in the graph by strengthening edges between semantically similar samples and weakening heterotrophic interference edges [17], [18], thus building a structural foundation better suited for GNNs propagation.

To improve GNN performance on heterophilic graphs, recent rewiring methods leverage neighborhood distributions [12]. Neighborhood Label Distribution (NLD) measures the class label proportions in a node's neighborhood, while Neighborhood Feature Distribution (NFD) captures the distribution of neighbor features. Comparing NLD and NFD allows the selective removal of edges connecting dissimilar nodes and the addition of edges between nodes with consistent distributions, mitigating heterophilic effects. Node-pair consistency is quantified via cosine similarity, yielding a similarity matrix Sim:

$$\mathrm{Sim}_{ij} = \mathrm{sim}(\mathrm{NFD}_i, \mathrm{NFD}_j) + \mathrm{sim}(\mathrm{NLD}_i, \mathrm{NLD}_j), \quad (6)$$

$$\mathrm{NFD}_i = (\tilde{\mathbf{A}})^K \cdot \mathbf{X}, \quad (7)$$

$$\mathrm{NLD}_i = \begin{cases} \dfrac{\sum_{v_j \in (N_i \cap V_{\text{train}})} Y_j^{\text{onehot}}}{|N_i \cap V_{\text{train}}|}, & \text{If } N_i \cap V_{\text{train}} \neq \emptyset \\ 0, & \text{Else} \end{cases}, \quad (8)$$

where $sim(u, v) = \frac{u^T v}{||u||_2 ||v||_2}$, $u$ and $v$ represent the embedding representations of nodes, respectively, and $||\cdot||_2$ denotes the Euclidean norm of a vector. This definition can measure the consistency of two nodes in direction, and is independent of the scale of the vector, making it more suitable for similarity calculation in high-dimensional embedding spaces.

We then rewire the original adjacency matrix $\mathbf{A}_0$ based on a threshold $\epsilon$. Specifically, edges with weights greater than $\epsilon$ are added to a set of new connections $E_{add}$, while those with weights below $\epsilon$ are removed and added to a set of dropped connections $E_{drop}$. The optimal value of $\epsilon$ was determined empirically via grid search during experimentation. We then combine $E_{add}$, $E_{drop}$, and the original edge set $E_0$ to synthesize the final edge set $E_{final}$, from which the

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rewired adjacency matrix $\mathbf{A}_1$ is constructed. By strengthening connections among nodes of the same class and preserving the original graph topology, this rewiring process enhances the GNN's ability to capture local class structures, thereby providing a more effective foundation for prototype contrastive alignment.

*2) JOSFC :* In this subsection, we introduce the JOSFC module to enhance model robustness through spectral resonance. The core idea is to align the principal subspaces of node features and graph structure by minimizing the angle between them in spectral space. Concurrently, we suppress the energy in their orthogonal directions, effectively denoising both the graph topology and node features. This ensures that subsequent GNN layers operate on cleaner, more consistent inputs. Intuitively, aligning the feature principal components with the graph's low-frequency spectral components promotes information propagation along semantically coherent paths. Therefore, our optimization objective enforces consistency by maximizing alignment between the feature subspace and the structural subspace [19], [20].

We propose a spectral resonance mechanism: spectral decomposition is applied to the normalized Laplacian $L$ and SVD to the feature matrix. Each spectral mode's alignment with class signals is assessed, and a spectral filter amplifies modes correlated with class information. The weighted adjacency matrix is sparsified and normalized for GNN message passing, enhancing class-discriminative components. Lanczos or Chebyshev approximations can reduce computational cost.

Specifically, we compute the spectral decompositions of $\mathbf{A}$ and $\mathbf{X}$ as:

$$\mathbf{A} = \mathbf{Q} \wedge \mathbf{Q}^T \quad \text{where} \quad \mathbf{Q} = (q_1, q_2, \ldots, q_n), \tag{9}$$

$$\mathbf{X} = \mathbf{U}\Sigma\mathbf{Z}^T \quad \text{where} \quad \mathbf{U} = (u_1, u_2, \ldots, u_n). \tag{10}$$

Then, we interpolate between the L largest eigenvectors to improve the alignment:

$$q_i = (1-\lambda_A)+\lambda_a \cdot \text{sign}(\langle q_i, u_j\rangle)u_j, \quad j = \arg\max_j |\langle q_i, u_j\rangle|, \tag{11}$$

$$u_t = (1-\lambda_X)+\lambda_X \cdot \text{sign}(\langle u_t, q_j\rangle)q_j, \quad j = \arg\max_j |\langle u_t, q_j\rangle|. \tag{12}$$

In formulas (11) and (12), we employ interpolation to fuse the local similarity graph with the globally distributed-based graph. This interpolation statistically helps dilute local noise and balance the frequency spectrum of graph signals in the spectral domain, thereby simultaneously preserving local information and global class structure.

After that, based on the new eigenvectors, we synthesize a new graph:

$$\widehat{\mathbf{A}} = \mathbf{Q}_{(i)} \wedge \mathbf{Q}_{(i)}^T, \tag{13}$$

$$\widetilde{\mathbf{X}} = \mathbf{U}_{(i)}\Sigma\mathbf{Z}_{(i)}^T. \tag{14}$$

The three steps of the JOSFC module are iterated for a fixed number of times, denoted by the hyperparameter k, which serves as the stopping criterion. In the end, we can obtain the final weighted dense graph: $\mathbf{A}_2 = \widetilde{\mathbf{A}}, \quad \mathbf{X}' = \widetilde{\mathbf{X}}$.

*3) Propagation and transform :* After ERSC and JOSFC complete their operations, propagation and transform begin on the dual channel. Both channels perform propagation using the denoised feature: $\mathbf{X}'$, generated by JOSFC. For example, in the ERSC channel, propagation uses a topic-sensitive PageRank algorithm, formalized as:

$$\begin{aligned} \mathbf{X}^{(0)} &= \mathbf{X}', \\ \mathbf{X}^{(t+1)} &= (1-\alpha)\widetilde{\mathbf{A}}_1\mathbf{X}^{(t)} + \alpha\mathbf{X}^{(0)}, \\ \mathbf{X}_1 &= \mathbf{X}^{(T)}, \end{aligned} \tag{15}$$

where $\alpha$ is the restart probability, and $\alpha \in (0, 1]$, $\widetilde{\mathbf{A}}_1$ is the normalized adjacency matrix of $\mathbf{A}_1$, $\mathbf{X}_1$ is the propagated feature matrix, T is the iteration number of propagation , and $t \in [0, T-1]$.

After propagation, we use the MLP as the classifer to transform $\mathbf{X}_1$ into the predicted label matrix $\hat{\mathbf{Y}}_1$ :

$$\mathbf{H}_1 = \text{MLP}_1(\mathbf{X}_1) = \delta(\mathbf{X}_1\mathbf{W}_1), \tag{16}$$

$$\hat{\mathbf{Y}}_1 = \text{MLP}_2(\mathbf{H}_1) = \text{softmax}(\mathbf{H}_1\mathbf{W}_2), \tag{17}$$

where $\mathbf{H}_1$ is the representation matrix transformed by $MLP_1$, and $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable parameter matrices. We then define the training loss as:

$$L_{ce} = -\frac{1}{|V^L|}\sum_{i\in V^L}\sum_{c=1}^{C} Y_{i,c}\log(\widehat{Y}_{1_{i,c}}), \tag{18}$$

where $\mathbf{y}_i$ is the true label of the i-th node, $\hat{\mathbf{y}}_i = softmax(\mathbf{x}_i^1)$.

The propagation and transformation process of JOSFC channel is consistent with that of ERSC channel and the loss function is:

$$L_{ce}^{'} = -\frac{1}{|V^L|}\sum_{i\in V^L}\sum_{c=1}^{C} Y_{i,c}\log(\widehat{Y}_{2_{i,c}}), \tag{19}$$

where $\mathbf{y}_i$ is the true label of the i-th node, $\hat{\mathbf{y}}_i^{'} = softmax(\mathbf{x}_i^2)$.

## C. CPA

This section details our improved CPA mechanism. Building upon the prototype alignment concept from D2PT [19], we have refined and simplified its architecture to seamlessly integrate with STEAM's dual-graph channel design. During each training iteration, we generate a distinct class prototype for each channel. This is achieved by computing a confidence-weighted average of node representations within each class, where the weights are derived from the nodes' predicted class probabilities. Formally, the prototype for class c is defined as:

$$\mathbf{P}_j^{(1)} = \frac{1}{CNF_j}\sum_{i\in C_j} cnf_i \cdot \mathbf{h}_i^{(1)}, \mathbf{P}_j^{(2)} = \frac{1}{CNF_j}\sum_{i\in C_j} cnf_i \cdot \mathbf{h}_i^{(2)}, \tag{20}$$

where $h_i^{(1)}$ and $h_i^{(2)}$ denote the representations of node i in the two channels, $cnf_i$ is the softmax confidence of node i belonging to class j, and $CNF_j = \sum_{i\in C_j} cnf_i$ is a normalization factor.

To enhance the semantic consistency between the dual-channel representations, we employ a prototype-level contrastive loss. This objective function encourages prototypes of the same class from the two channels to align closely in the embedding space, while simultaneously pushing apart

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prototypes of different classes [19], [21]. In contrast to the more complex InfoNCE [19] loss, we adopt a simpler and more efficient formulation based on temperature-scaled cosine similarity:

$$\mathcal{L}_{\text{CPA}} = -\sum_{j=1}^{c} \log \frac{\exp\left(\frac{sim(p_j^{(1)}, p_j^{(2)})}{\tau}\right)}{\sum_{k=1}^{c} \exp\left(\frac{sim(p_j^{(1)}, p_k^{(2)})}{\tau}\right)}, \tag{21}$$

where $sim(\cdot,\cdot)$ denotes the cosine similarity and $\tau$ is the temperature parameter.

Our ultimate learning objective is to combine the three loss functions mentioned above and assign weights $\gamma, \beta$ to $L_{ce}$ and $L_{ce}^{'}$ separately:

$$L = \gamma L_{ce} + \beta L_{ce}^{'} + L_{\text{CPA}}. \tag{22}$$

## IV. Experiments

In this section, we conduct extensive experiments to verify the effectiveness of STEAM, aiming to answer the following research questions:

**RQ1**: Does STEAM achieve better performance on the DGA task compared to other baseline methods?

**RQ2**: What is the contribution of each component in STEAM to the overall performance?

**RQ3**: How do hyperparameters affect the performance of STEAM?

**RQ4**: How robust is STEAM to measurement noise?

**RQ5**: Can STEAM learn a representation space with better inter-class separability and intra-class compactness?

### A. Experimental Setup

*1) Datasets Description :* We perform extensive experiments on two publicly available benchmark datasets, DGA2000 and DGA200. Detailed statistics of these two datasets are provided in Table I and Table II. The optimal number of neighbors $K$ for graph construction was determined for each dataset through validation, with the results being $K$=10 for DGA-2000 and $K$=4 for DGA-200.

**DGA-200**: The DGA-200 dataset contains 201 samples and covers 7 types of faults: partial discharge (PD), spark discharge (SD), arc discharge (AD), low-temperature overheating (LT), low-to-medium temperature overheating (LMT), medium-temperature overheating (MT), and high-temperature overheating (HT).

**DGA-2000**: The DGA-2000 dataset contains 1982 samples and includes 6 types of faults: low-temperature overheating (LT), medium-temperature overheating (MT), high-temperature overheating (HT), high-energy discharge (HE), low-energy discharge (LE), partial discharge (PD), as well as 1 normal type (Normal).

The samples in both datasets contain consistent dissolved gas information: hydrogen ($H_2$), methane ($CH_4$), ethane ($C_2H_6$), ethylene ($C_2H_4$), and acetylene ($C_2H_2$).

It is worth noting that the DGA-200 and DGA-2000 datasets have different class definitions. In this paper, our two datasets are completely independent, with separate model training and evaluation, and we have not attempted cross-dataset transfer or unified class labels. The STEAM framework is designed to be class-agnostic: graph construction, structure optimization modules (ERSC, JOSFC), and the classifier do not depend on a preset number of classes; the only module involving class information is Prototype Alignment (CPA), whose class number is dynamically obtained directly from the training labels of the current dataset. Therefore, STEAM can naturally adapt to datasets with different numbers of classes without modifying the model architecture. The experimental results on the two datasets are independent and comparable, ensuring the fairness and validity of the evaluation.

TABLE I
THE STATISTICAL INFORMATION OF DGA-200 AND DGA-2000

| Dataset | #Node | #Edge | #Class | #Features |
|---|---|---|---|---|
| DGA-200 | 201 | 804 | 7 | 5 |
| DGA-2000 | 1982 | 7928 | 7 | 5 |

TABLE II
THE STATISTICS OF FAULT TYPES IN DGA-200 AND DGA-2000 DATASETS

| Dataset | #SD | #LT | #AD | #HT | #MT | #LMT | #PD | Sum |
|---|---|---|---|---|---|---|---|---|
| DGA-200 | 49 | 19 | 54 | 38 | 9 | 16 | 16 | 201 |
| Dataset | #PD | #LE | #HE | #LT | #MT | #HT | #Normal | Sum |
| DGA-2000 | 113 | 276 | 224 | 111 | 113 | 438 | 707 | 1982 |

*2) Baselines and Evaluation Metrics:* To comprehensively evaluate the performance of STEAM on the DGA task, we compare it against several baselines, including:

**Traditional machine learning methods**: Support Vector Machine classification (SVM-C) [4], Partial Discharge Localization K-Nearest Neighbors (PDL-KNN) [16], and Interpretable Decision Trees (IntDTree) [22].

**Graph rewired methods**: DHGR [12], and JDR [13].

**GNNs-based methods**: GraphSmin [10], DGRCL [9].

To more precisely assess the performance of STEAM, we adopt commonly used evaluation metrics: Accuracy, AUC (Area Under the Curve), F1, Precision, and Recall. Since our task is a multi-class classification problem, the AUC metric is computed using macro-averaging, which treats all classes equally and ensures fair evaluation across classes.

*3) Hyperparameter Settings:* The STEAM model is implemented using PyTorch and trained with the Adam optimizer. Hyperparameters for STEAM and all baseline methods are selected via grid search with 5-fold cross-validation on the training set. Specifically, for each dataset, we first perform a stratified random split into 70% training+validation and 30% test sets, preserving the original class distribution. The test set is held out and used only for final evaluation. Within the 70% training+validation data, we apply 5-fold cross-validation: the data is partitioned into five folds, with four folds used for training and the remaining fold for validation in each iteration. The grid search covers the following ranges: learning rate $\{0.001, 0.005, 0.01, 0.05, 0.1\}$, dropout rate $\{0, 0.1, 0.2, 0.5\}$, hidden dimension $\{64, 128, 256\}$, number of layers

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TABLE III

PERFORMANCE COMPARISON (THE OPTIMAL RESULT IN **BOLD**, THE SUBOPTIMAL UNDERLINED); LAST ROW SHOWS P-VALUES OF STEAM VS. SUBOPTIMAL METHOD. THE REPORTED VALUES ARE THE MEAN ± STANDARD DEVIATION OF 30 INDEPENDENT STRATIFIED RANDOM PARTITIONS

| Method | DGA2000 | | | | | DGA200 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Accuracy | AUC | F1 | Precision | Recall | Accuracy | AUC | F1 | Precision | Recall |
| SVM-C | 70.03±0.01 | 90.16±0.04 | 53.35±0.01 | 52.18±0.03 | 54.57±0.01 | 65.85±0.02 | 87.29±0.04 | 57.90±0.08 | 59.73±0.05 | 56.17±0.04 |
| PDL-KNN | 71.54±0.03 | 91.27±0.03 | 60.37±0.03 | 61.80±0.06 | 59.01±0.05 | 70.73±0.07 | 83.55±0.08 | 58.82±0.07 | 57.14±0.02 | 60.60±0.06 |
| IntDTree | 71.24±0.02 | 86.10±0.05 | 57.21±0.04 | 57.59±0.02 | 56.84±0.03 | 68.29±0.06 | 82.27±0.07 | 57.26±0.03 | 56.50±0.03 | 58.04±0.08 |
| DHGR | 72.80±0.05 | 90.88±0.01 | 57.58±0.04 | 61.81±0.03 | 54.43±0.09 | 69.76±0.02 | 90.69±0.05 | 44.85±0.01 | 47.15±0.01 | 42.76±0.01 |
| JDR | 72.39±0.01 | 90.80±0.01 | 54.24±0.02 | 54.27±0.02 | 54.23±0.02 | 70.71±0.02 | 90.36±0.01 | 57.63±0.02 | 52.84±0.01 | 63.38±0.02 |
| GraphSmin | 76.46±0.05 | 90.45±0.04 | 76.35±0.09 | 76.63±0.03 | 76.04±0.02 | 74.64±0.02 | 93.17±0.05 | 75.36±0.08 | 77.48±0.06 | 73.34±0.06 |
| DGRCL | 75.36±0.02 | 91.25±0.01 | 78.12±0.02 | 79.87±0.01 | 76.44±0.01 | 73.43±0.05 | **94.32±0.02** | 81.62±0.06 | 82.76±0.05 | 80.57±0.04 |
| STEAM (ours) | **80.30±0.05** | **94.56±0.07** | **80.07±0.07** | **81.82±0.07** | **78.41±0.01** | **75.73±0.04** | 91.50±0.03 | **82.53±0.05** | **83.57±0.09** | **81.46±0.07** |
| p-value | $2.46\times10^{-8}$ | $3.01\times10^{-12}$ | $3.55\times10^{-9}$ | $2.56\times10^{-4}$ | $4.66\times10^{-8}$ | $7.44\times10^{-5}$ | $3.63\times10^{-4}$ | $2.04\times10^{-5}$ | $9.64\times10^{-9}$ | $7.16\times10^{-6}$ |

$\{2, 3, 4\}$, and for STEAM-specific parameters, number of neighbors $K \in \{1,2,3,4,5,6,7,8,9,10,11\}$ and similarity threshold $\epsilon \in \{0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9\}$. The combination achieving the highest validation accuracy is selected for final evaluation on the test set. All baseline methods undergo the same tuning procedure to ensure fair comparison.

To ensure statistical reliability, the stratified splitting is repeated 30 times using different random seeds, producing 30 independent train/test splits. All models are evaluated on the same splits, and the reported results correspond to the mean performance over the 30 runs, with standard deviations indicating variability.

## B. Performance Comparison (RQ1)

To evaluate the efficacy of STEAM, we conduct experiments on both datasets, comparing it with the aforementioned baselines using identical data splits. Table III summarizes performance across five key metrics. To confirm that the observed improvements are statistically significant, we performed t-tests comparing STEAM with the suboptimal method for each metric. As reported in the last row of Table III, all p-values are below 0.05, indicating that STEAM's performance gains are highly significant.

To ensure fair comparisons, we reimplemented all baseline methods according to the descriptions in the original literature and performed hyperparameter optimization and evaluation on the DGA-200 and DGA-2000 datasets using the exact same training/validation/test partitions as STEAM. Therefore, while the original experiments in the references may have targeted different datasets or numbers of classes, all methods in this study were retrained under uniform conditions.

On DGA2000, STEAM achieves 80.30% accuracy, outperforming traditional baselines such as SVM-C (70.03%), PDL-KNN (71.54%), IntDTree (71.24%), and advanced rewiring methods DHGR (72.80%) and JDR (72.39%). It also attains the highest AUC (94.56%), F1-score (80.07%), Precision (81.82%), and Recall (78.41%), with Recall exceeding the nearest competitor DGRCL by 1.97 points, demonstrating strong capability in identifying hard-to-classify samples.

On DGA200, which has lower data quality, STEAM maintains superior performance, achieving 75.73% accuracy, 91.50% AUC, 82.53% F1-score, 83.57% Precision, and 81.46% Recall, confirming its robustness under challenging conditions.

Overall, STEAM consistently enhances performance across datasets, effectively recognizing difficult samples, distinguishing fault types, and maintaining balanced metrics. These results validate its effectiveness for imbalanced and weakly supervised graph tasks, and more importantly, demonstrate that STEAM provides a general graph representation learning paradigm for industrial time-series or tabular data with weak graph structures and ambiguous categories, beyond the DGA scenario.

## C. Ablation Study (RQ2)

To verify the contribution of each module in STEAM, we conduct ablation experiments by progressively removing or replacing key components of the model, including:

**w/o ERSC**: Without the ERSC module.

**w/o JOSFC**: Without the JOSFC module.

**w/o CPA**: Without the CPA module.

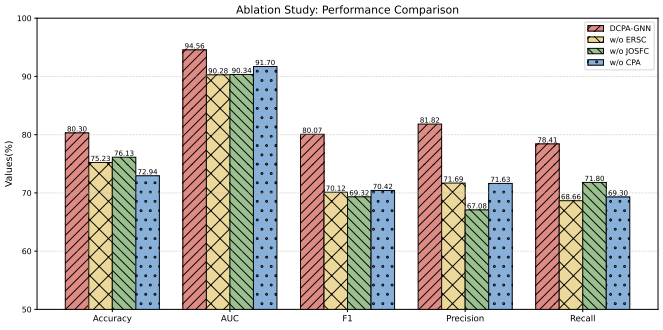



Fig. 3. Ablation Study Results on DGA-2000.

From Figure 3, the ablation study yields the following conclusions:

(1) Removing the Edge-aware Relational Structure Consistency (ERSC) module—which omits neighborhood feature/label distributions and direct node-pair similarity propagation—resulted in significant performance degradation: Accuracy decreased by 5.07%, AUC by 4.28%, F1 by 9.95%, Precision by 10.13%, and Recall by 9.75%. The pronounced

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drop in AUC indicates that ERSC critically enhances the model's ranking capability across classes, thereby improving generalization. The overall substantial decline across metrics suggests that ERSC effectively improves sample coverage, reduces missed detections, and strengthens the model's ability to capture edge samples.

(2) Without the Joint Orthogonal Structure-Feature Consistency (JOSFC) module, which enforces consistency in spectral space and aligns feature-structure subspaces along principal directions, performance dropped as follows: Accuracy by 4.17%, AUC by 4.22%, F1 by 10.75%, Precision by 14.74%, and Recall by 6.61%. The sharp decline in F1-score highlights the importance of JOSFC in addressing inter-class imbalance and ambiguous decision boundaries. Meanwhile, the significant drop in Precision indicates that JOSFC substantially reduces false positives, likely due to more accurate structural modeling and subspace alignment.

(3) Removing the Cross-Channel Prototype Alignment (CPA) module, which performs class-wise alignment across channels, led to the broadest performance decrease across all metrics: Accuracy fell by 7.36%, AUC by 2.86%, F1 by 9.65%, Precision by 10.19%, and Recall by 9.11%. The severe decline in accuracy underscores CPA's central role in enhancing global discriminative power. It effectively aligns and stabilizes classification boundaries, thereby strengthening inter-class separability and overall model consistency.

In summary, all three proposed modules contribute significantly to model performance. The ablation results confirm that each component has a distinct and essential function: ERSC improves generalization and sample coverage, JOSFC enhances structural discriminability and reduces false positives, and CPA ensures class-wise consistency and global separation. The complete STEAM framework integrates these complementary strengths, which is necessary for achieving accurate and robust detection performance.

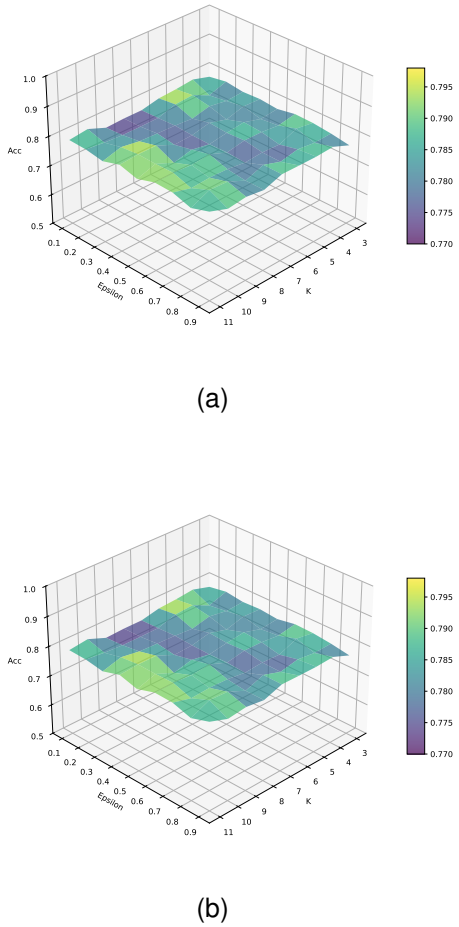



Fig. 4. Hyperparameter analysis on the DGA-2000 dataset. (a) Accuracy and (b) F1-score as 3D surfaces as functions of the number of neighbors $K$ and similarity threshold $\epsilon$.

## D. Hyperparameter Analysis (RQ3)

To delve into the impact of key hyperparameters—the number of neighbors $K$ and the similarity threshold $\epsilon$—on model performance, we performed a grid search on the DGA-2000 dataset and presented the changes in accuracy (ACC) and F1 score as a function of ($K$, $\epsilon$) as 3D surface plots, as shown in Figure 4.

Figure 4(a) shows that the ACC surface is generally flat in parameter space, with values stable between 0.77 and 0.80. The highest point ($K$=10, $\epsilon$=0.8) reaches 0.803. Even when the parameter combinations deviate from the optimal region (e.g., $K$=4, $\epsilon$=0.3), the ACC remains above 0.77, with fluctuations less than 3%. This phenomenon indicates that STEAM is insensitive to the values of $K$ and $\epsilon$, maintaining stable classification accuracy over a wide parameter range.

Figure 4(b) shows the surface of the F1 score, whose trend is highly consistent with that of the ACC. The F1 value generally ranges from 0.74 to 0.80, with the highest point near $K$=9, $\epsilon$=0.7 (F1=0.8007). The surface also exhibits a flat plateau characteristic. This indicates that STEAM also has good robustness in maintaining a balance between precision and recall.

In summary, the smooth characteristics of the ACC and F1 surfaces reveal two important advantages of STEAM: (1) parameter tuning friendliness: users can obtain near-optimal performance over a wide parameter range without performing fine-grained grid searches, reducing the reliance on expert experience in actual deployments; (2) strong robustness: the model can still stably output reliable diagnostic results when faced with fluctuations in data distribution or parameter estimation errors. These characteristics further confirm the practical value of STEAM in actual transformer monitoring scenarios.

## E. Noise robustness analysis (RQ4)

To evaluate the robustness of STEAM against real-world measurement uncertainties, we combine methodological elements from Ahmad & Sanaye-Pasand [23] and Bouzar et al. [24]. From Ahmadi & Sanaye-Pasand, we adopt the concept of adding random uncertainties within a specified relative error bound. From Bouzar et al., we adopt the practice of independently randomizing the perturbed values across different gases to avoid artificially preserving gas ratios, which would otherwise hide measurement errors.

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TABLE IV

ACCURACY (%) UNDER DIFFERENT RELATIVE MEASUREMENT UNCERTAINTY LEVELS ON DGA-2000 AND DGA-200 DATASETS.

| Dataset | Method | $r = 0$ | $r = \pm 5\%$ | $r = \pm 10\%$ | $r = \pm 15\%$ | $r = \pm 20\%$ |
|---|---|---|---|---|---|---|
| DGA-2000 | GCN | 74.51 ± 0.04 | 72.86 ± 0.05 | 70.23 ± 0.07 | 66.08 ± 0.08 | 64.47 ± 0.10 |
| | DHGR | 72.81 ± 0.03 | 71.24 ± 0.04 | 68.95 ± 0.06 | 65.22 ± 0.07 | 64.81 ± 0.09 |
| | JDR | 72.42 ± 0.04 | 70.68 ± 0.05 | 68.24 ± 0.06 | 64.35 ± 0.08 | 63.76 ± 0.10 |
| | DGCRL | 75.45 ± 0.03 | 75.03 ± 0.05 | 74.89 ± 0.06 | 72.54 ± 0.07 | 69.12 ± 0.09 |
| | **STEAM** | **80.30 ± 0.05** | **79.54 ± 0.04** | **78.32 ± 0.05** | **76.18 ± 0.06** | **73.14 ± 0.07** |
| DGA-200 | GCN | 70.02 ± 0.05 | 68.01 ± 0.06 | 65.24 ± 0.08 | 61.33 ± 0.09 | 56.02 ± 0.11 |
| | DHGR | 69.84 ± 0.04 | 67.82 ± 0.05 | 65.08 ± 0.07 | 60.95 ± 0.08 | 55.47 ± 0.10 |
| | JDR | 70.78 ± 0.08 | 68.64 ± 0.06 | 65.97 ± 0.07 | 61.78 ± 0.09 | 56.08 ± 0.11 |
| | DGCRL | 73.46 ± 0.04 | 72.85 ± 0.07 | 71.42 ± 0.08 | 68.97 ± 0.09 | 67.23 ± 0.10 |
| | **STEAM** | **75.73 ± 0.04** | **74.91 ± 0.05** | **70.68 ± 0.06** | **71.49 ± 0.07** | **68.25 ± 0.08** |

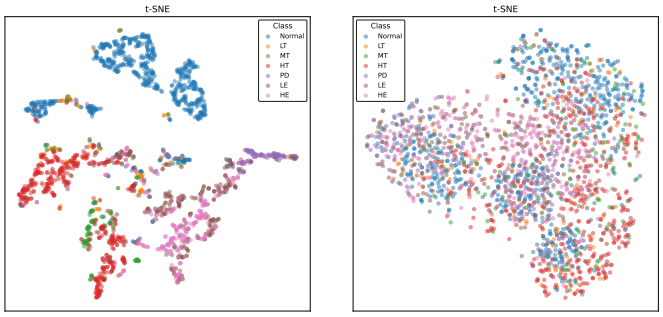
Following the discussion in Bouzar et al., we evaluated five settings: a clean baseline (r=0) and four uncertainty levels corresponding to relative error bounds of ±5%, ±10%, ±15%, and ±20%. For each gas concentration, we generated uniformly distributed values within its uncertainty interval. The sequences of generated values for different gases were then independently and randomly shuffled. The generated values were used to reconstruct the k-NN graph with the same K values as in the reference experiments (K=4 for DGA-200, K=10 for DGA-2000).

What's more, in this study, robustness is evaluated under a protocol where models are retrained and tested at the same noise level, reflecting realistic deployment conditions in which historical data inherently contain measurement uncertainties from sampling, gas extraction, chromatographic analysis, and operational fluctuations. Training on noisy data thus simulates learning from imperfect records in practice.For graph-based methods, this setting also ensures consistency in graph construction between training and testing, avoiding confounding effects caused by topology mismatch. Although training on clean data and testing on noisy data provides an alternative way to assess inference-stage robustness, it may introduce inconsistencies in graph structure that obscure the evaluation.Therefore, the adopted protocol offers a more reliable assessment of robustness under practical conditions. The results show in Table IV that STEAM maintains superior performance in this setting.

As shown in Table IV, increasing measurement uncertainty degrades all methods, but STEAM maintains the best overall performance. On DGA-2000, STEAM drops from 80.30% (clean) to 73.14% at ±20% uncertainty, outperforming DGCRL (69.12%) by 4.02 points. On DGA-200, STEAM reaches 68.25% vs. DGCRL's 67.23% (+1.02 points). The performance gap widens as uncertainty increases, indicating better robustness. DGCRL declines by 6.33 points on DGA-2000 (75.45% to 69.12%) and 6.23 points on DGA-200 (73.46% to 67.23%); STEAM declines by 7.16 and 7.48 points respectively. Despite slightly steeper declines, STEAM's higher initial accuracy ensures superior final accuracy under all noise levels, demonstrating its overall effectiveness.

### *F. Visualization (RQ5)*

To rigorously assess STEAM's capability in mitigating blurred class boundaries, we conduct t-SNE–based visualization on a randomly sampled subset of the DGA-2000 test set. As depicted in Fig. 5, the original input features (Fig. 5(a)) exhibit substantial inter-class entanglement and indistinct decision boundaries. In contrast, the representations learned by STEAM (Fig. 5(b)) form markedly tighter intra-class manifolds and exhibit clearer inter-class separations. These results indicate that STEAM, enabled by its graph rewiring strategy and CPA module, substantially enhances the structural and semantic coherence of feature representations, thereby yielding more discriminative embeddings and effectively alleviating the blurred boundary issue in DGA-based transformer fault diagnosis.




(a) (b)

Fig. 5. t-SNE Visualization on DGA-2000. (a) Original Features. (b) Features after learning.

## V. DISCUSSION

While hyperparameter tuning for each dataset is a standard practice in machine learning research to ensure fair comparisons, the practical deployment of diagnostic models in industrial settings raises important considerations regarding usability and data dependence. This section discusses how STEAM addresses these concerns.

STEAM introduces two core hyperparameters—the number of neighbors K and the similarity threshold —which possess clear physical interpretations and are limited in number. In a real-world deployment scenario, these parameters can be optimized offline using historical data, after which the model is fixed. Subsequently, end-users—typically maintenance personnel—only need to provide new gas concentration measurements to receive fault diagnoses, without any further tuning or technical intervention.

The model's robustness to data variations is evidenced by its core design and empirical results. The modules ERSC, JOSFC, and CPA operate on intrinsic data structures—neighborhood distributions, spectral components, and class prototypes—rather than relying on fixed statistical assumptions. This enables automatic adaptation to different class numbers, sample sizes, and noise levels. Moreover, the consistent state-of-the-art performance on two distinct datasets (DGA-200 and DGA-2000) with different class definitions and distributions confirms the model's generalization capability, rather than overfitting to specific data characteristics.

Future research will explore more automated hyperparameter selection mechanisms—such as Bayesian optimization, AutoML, or meta-learning—to further reduce manual effort and

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enhance accessibility for industrial practitioners. Additionally, developing heuristic rules based on data statistics and graphical user interfaces could facilitate adoption by non-expert users.

## VI. Conclusion and Future Work

In this work, we propose STEAM a novel graph-based model that addresses two critical challenges in power transformer fault detection: weak graph quality and high inter-class similarity. To this end, we design three dedicated modules: ERSC to enhance local structure by leveraging neighborhood consistency, JOSFC to jointly refine structural and feature representations for better boundary clarity and suppress noise, and CPA to align class-wise distributions across dual channels, thereby improving inter-class separability. These components work collaboratively to improve representation quality, leading to high fault detection accuracy. STEAM offers a new approach to graph modeling and representation learning for power transformer fault diagnosis. In future work, we will explore its applicability to other industrial fault diagnosis tasks and attempt to incorporate more efficient graph structure learning mechanisms to improve the model's scalability and real-time capability.

Although STEAM is designed and validated based on DGA data, its architecture is data-independent: it transforms tabular input into a graph and refines it using modules (ERSC, JOSFC, CPA) based on inherent data properties (neighborhood distribution, spectral components, and class prototypes) rather than domain-specific heuristics. This allows the framework to be broadly applicable to other transformer health indicators (e.g., oil quality indicators, partial discharge signals) and pervasive industrial diagnostic tasks such as noise, class imbalance, and boundary ambiguity. Consistent performance across two different DGA datasets and various noise levels initially demonstrates its generalization ability. However, further validation is needed. Therefore, future work will evaluate its cross-dataset generalization ability by training on one dataset (e.g., DGA-2000) and testing on another dataset (e.g., IEC TC-10) without readjustment, explore domain adaptation to handle distribution variations, and extend the framework to other sensor modalities to determine its versatility in real-world monitoring scenarios.

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